EXP.NO:07 Flow control at Data link layer (Sliding Window Protocol)

DATE: 24/9/26

AIM:

Write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL. Simulate the flow of frames from one node to another.

PROGRAM:

Sender.py:

```
import time
sender_file = "Sender_Buffer.txt"
receiver_file = "Receiver_Buffer.txt"
window_size = int(input("Enter window size : "))
message = input("Enter Text Message : ")
total_frames = len(message)
start = 0
print("\nMessage :", message)
print("Total Frames:", total_frames)
while start < total_frames:
    end = start + window_size
    if end > total_frames:
        end = total_frames
    file = open(sender_file, "w")
    print("\nSending Frames:")
```

```
        for i in range(start, end):
            file.write(str(i)+","+message[i]+"\n")
            print("Frame",i,"DATA=",message[i])
        file.close()
        print("\nSender_Buffer.txt updated.")
        print("Waiting for Receiver...")
        time.sleep(3)
        file = open(receiver_file,"r")
        response = file.read()
        file.close()
        data = response.split(":")
        response_type = data[0]
        response_number = int(data[1])
        if response_type == "ACK":
            print("\nACK received:",
response_number)
            start = response_number
            print("Window slides to
Frame", start)
        elif response_type == "NACK":
            print("\nNACK received:",
response_number)
            print("Resending frames from Frame",
response_number)
            start = response_number
            time.sleep(2)
print("\n----------")
print("Transmission completed")
print("----------")
```

receiver.py

```
import time
sender_file = "Sender_Buffer.txt"
receiver_file = "Receiver_Buffer.txt"
file = open(sender_file,"w")
file.close()
expected = 0
print("----------")
print("Receiver started")
print("----------")
while True:
    file = open(sender_file,"r")
    lines = file.readlines()
    file.close()
    if len(lines) == 0:
        time.sleep(1)
        continue
    error = False
    print("\nReading Sender_Buffer.txt")
```

```
        for line in lines:
            data = line.strip().split(",")
            frame_no = int(data[0])
            character = data[1]
            print("Frame", frame_no,
"DATA =", character)

            if frame_no != expected:
                print("\nFrame error!")
                print("Expected Frame:",
expected)
                print("Received Frame:",
frame_no)
                file = open(receiver_file, "w")
                file.write("NACK:" +
str(expected))
                file.close()
                print("NACK sent:",
expected)
                error = True
                break
            expected = expected + 1
        file = open(sender_file, "w")
        file.write("ACK:" + str(expected))
        file.close()
        print("ACK Sent:", expected)
        time.sleep(1)
```

Output:

```
Sender.py
Enter Window size: 3
Enter Text Message: Hello
Message: Hello
Total Frames: 5

Sending Frames:
Frame 0 DATA = H
Frame 1 DATA = e
Frame 2 DATA = l
Sender_Buffer.txt updated.
Waiting for the Receiver...
ACK received: 3
Window slides to Frame 3

Sending Frames:
Frame 3 DATA = l
Frame 4 DATA = o
Sender_Buffer.txt updated.
Waiting for the Receiver...
ACK received: 5
Window slides to Frame 5
----------
Transmission completed
----------
```

receiver.py

```
- - - - - - - - - - - -
Receiver started
- - - - - - - - - - - -
Reading Sender_Buffer.txt
Frame 0 DATA = H
Frame 1 DATA = e
Frame 2 DATA = l
ACK sent : 3

Reading Sender_Buffer.txt
Frame 3 DATA = l
Frame 4 DATA = o
ACK sent : 5
```

